

Double Gamma Principal Components Analysis

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Abstract

This paper proposes a Double Gamma (DGamma) Principal Components Analysis (DGamma PCA); it considers effective PCA method to noise. We will utilize the DGamma distribution to model noise. An exact form of the probability density function (pdf) of DGamma distribution will be viewed. In addition, introduce some of graphical illustration of the pdf of the DGamma distribution; view Moment generation function of DGamma distribution. Moreover, maximum likelihood estimator (MLE) of DGamma distribution is obtained. Finally, experimental results on simulated data of DGamma PCA to noise are demonstrated.

Keywords: Double Gamma distribution, Maximum likelihood estimation, DGamma PCA

1. Introduction

The dimension reduction is a process of projecting high-dimensional data to a much lower-dimensional space. Determining patterns in data of high dimension can be hard to find. Principal Components Analysis (PCA) is a way of identifying patterns in data, and expressing the data so that PCA is considering a powerful tool for analyzing data [1]. PCA based on Gaussian noise model is sensitive to the noise [2].

Principal Components Analysis (PCA) is a standard statistical tool which have been widely used in dimensionality reduction, data compression and image processing. It looks for a linear transformation where reduces a large set of variables in which the amount of variance in the data is maximal.

PCA method is applied in many fields, such as pattern recognition [3], image processing [4], regression application [5] and data mining [6].

Historically, a number of natural approaches of PCA have been explored and proposed in the literature over several decades. Robust PCA methods can be categorized into two paradigms: non-probabilistic approaches and probabilistic approaches. The basic strategy of non-probabilistic methods is to remove the influence of large noise in corrupted data items. While, probabilistic approaches demonstrating that PCA may indeed be derived within a density-estimation framework [7].

Actually, noise in data reduces the quality of the information. PCA is considered one of the techniques that is interested in reducing the number of dimensions and extract the most important information without much loss of information [8]. Many of studies of PCA assume that the data are distributed according to a Gaussian distribution. The Gaussian PCA is sensitive to the noise of large magnitude. To robustify PCA, a number of improvements have been proposed by replacing the Gaussian distribution by another one [9]. The objective of this paper is replacement the Gaussian PCA to the DGamma PCA. A new approach of DGamma PCA of modeling noise is studied and the results are obtained.

This paper is organized as follows: Section 2 presents DGamma distribution: it's probability density function, graphical illustration, Moment generating function and the computation of the maximum likelihood estimation. Section 3 presents the maximum likelihood estimation of the parameters of the DGamma distribution. Then, Section 4 provides case study of a simulated DGamma PCA of modeling noise. Finally, conclusions are drawn in Section 5.

2. Double Gamma Distribution

The Gamma distribution (or also known as the Erlang distribution, named for the Danish mathematician Agner Erlang) has considerable attention in reliability theory [10]. The general form for the probability density function (pdf) of the DGamma distribution (also referred to as reflected gamma distribution) is given by

$$f(x; \mu, \theta_1, \theta_2) = \left(\frac{1}{2}\right) \frac{\left(\frac{|x-\mu|}{\theta_2}\right)^{(\theta_1-1)} e^{-\left(\frac{|x-\mu|}{\theta_2}\right)}}{\theta_2 \Gamma(\theta_1)}, \quad -\infty < x < \infty, \quad (1)$$

where μ and θ_2 are the positive location and scale parameters respectively, and Γ is the gamma function which has the form

$$\Gamma a = \int_0^{\infty} t^{a-1} e^{-t} dt.$$

The form of the DGamma distribution when $\mu = 0$ is given by

$$f(x; \mu, \theta_1, \theta_2) = \left(\frac{1}{2}\right) \frac{\left(\frac{|x|}{\theta_2}\right)^{(\theta_1-1)} e^{-\left(\frac{|x|}{\theta_2}\right)}}{\theta_2 \Gamma(\theta_1)}. \quad (2)$$

The standard forms of DGamma distribution of equation (1) where $\mu = 0$ and $\theta_2 = 1$ is given by

$$f(x; \theta) = \left(\frac{1}{2}\right) \frac{|x|^{(\theta-1)} e^{-|x|}}{\Gamma(\theta)}. \quad (3)$$

Some different shapes of pdf for DGamma distribution in different values of their parameters are presented. The following is the plot of the DGamma probability density function

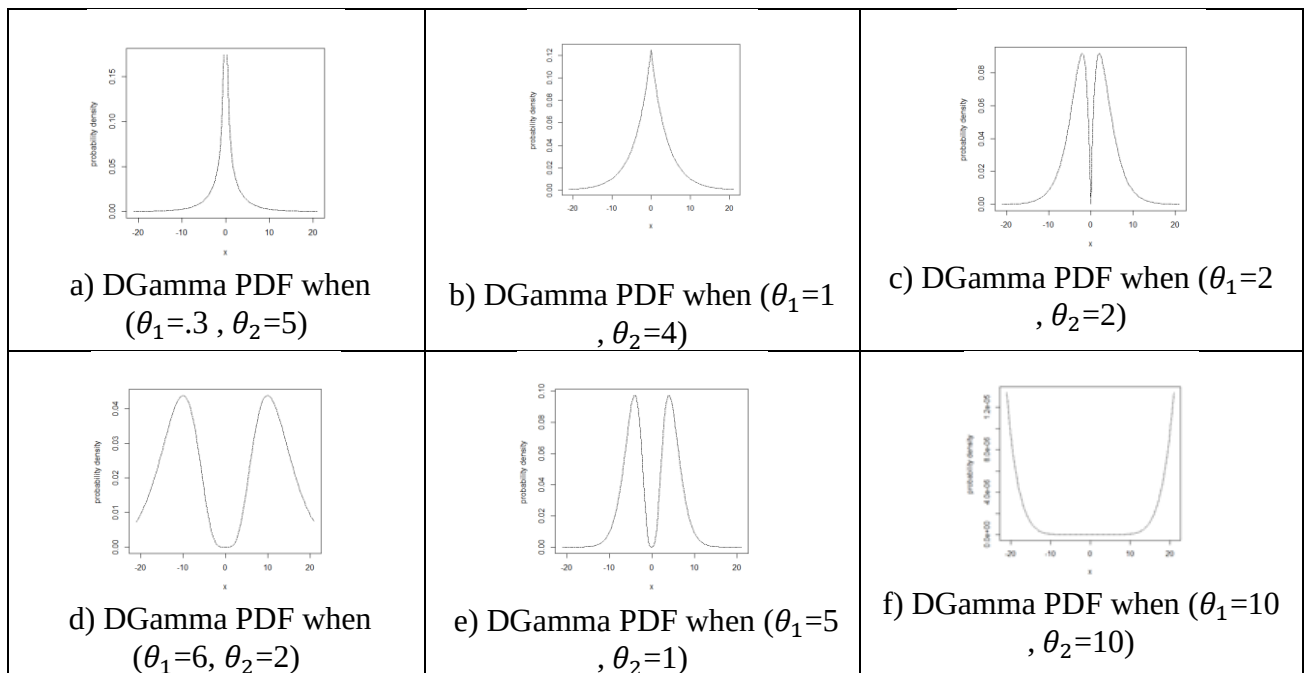


Figure 1: Different shapes of the DGamma densities.

Figure 1 shows different shapes of DGamma densities. Where plots [c], [d] and [e] show clearly the densities are bimodal, with a valley spreading them. The density in plot [f] is a bathtub shape. Whereas in plot [b], when $\theta_1 = 1$, the density shape reduces to the Laplace distribution (non-smoothness property). Finally in plot [a] the pdf curve appears like two exponential distributions one increasing and the other is decreasing where $\theta_1 < 1$.

The moment generating function of DGamma in equation (3) is:

$$M(t) = \frac{1}{2(1-t)^\theta} + \frac{1}{2(1+t)^\theta}.$$

3. Maximum Likelihood Estimation

Definition: Let represent $x_1, x_2, \dots, x_n$ a random sample from a density function $f(x; \theta)$ and let $L(\theta) = L(\theta; x_1, x_2, \dots, x_n)$ be corresponding likelihood function, and is given by:

$$L(\theta) = L(\theta; x_1, x_2, \dots, x_n) = \prod_{i=1}^n f(x_i; \theta).$$

As a general procedure for constructing estimators, the value of $\hat{\theta}$, that maximize $L(\theta)$ will be chosen. Any value of $\hat{\theta}$ satisfies the following inequality

$$L(\hat{\theta}) \geq L(\theta) \text{ for all } \theta \neq \hat{\theta}. \quad [11]$$

The likelihood function for n i.i.d observations $(x_1, \dots, x_n)$ from DGamma with $\mu = 0$ is

$$(2\theta_2 r \theta_1)^{-n} \left(\prod_{i=1}^n \frac{|x_i|}{\theta_2} \right)^{\theta_1 - 1} e^{-\sum_{i=1}^n \frac{|x_i|}{\theta_2}},$$

giving the log-likelihood:

$$-n \ln(2\theta_2 r \theta_1) + (\theta_1 - 1) \sum_{i=1}^n \ln \left(\frac{|x_i|}{\theta_2} \right) - \sum_{i=1}^n \frac{|x_i|}{\theta_2}. \quad (4)$$

Hence, a numerical matter can be used to solve the log-likelihood Equation (4) in order to compute the maximum likelihood estimates of θ_1, θ_2 .

The maximum likelihood estimates of the two parameters of the DGamma distribution is applied for different random samples of size n generated from DGamma distribution. Then, the function “nlm” from package “STATS4” of R statistical package is used to compute the ML estimates of $\hat{\theta}_1$ and $\hat{\theta}_2$. The confidence intervals, MSE and Bias are also computed.

n,R	θ_1	θ_2	$\hat{\theta}_1$	$\hat{\theta}_2$	Confidence interval of $\hat{\theta}_1(L, U)$	Confidence interval $\hat{\theta}_2(L, U)$	MSE of $\hat{\theta}_1$	MSE of $\hat{\theta}_2$	Bias of $\hat{\theta}_1$	Bias of $\hat{\theta}_2$
24,50	5	2	7.257404	1.860666	(5.235087, 9.279721)	(0.4382013, 3.2831313)	0.2123280457	0.0008089117	0.094058503	0.005805571
	2.5	7	3.252194	6.738703	(0.8550092, 5.6493796)	(4.645845, 8.831560)	0.023574851	0.002844845	0.03134143	0.01088739
	.9	.05	1.25774564	0.04234315	(0.6838629, 1.8316283)	(0.01839678, 0.06628953)	5.332581e-03	2.442804e-06	0.0149060684	0.0003190352
	.5	.3	0.5824239	0.2891483	(0.1835674, 0.9812804)	(0.1351743, 0.4431222)	2.830707e-04	4.906676e-06	0.0034343285	0.0004521558
50,25	5	2	5.668794	1.931147	(3.291254, 8.046335)	(1.333724, 2.528569)	8.945720e-03	9.481596e-05	0.0178454	0.002338958
	2.5	7	2.610725	6.858722	(1.471981, 3.749468)	(4.599197, 9.118247)	0.0002451999	0.0003991898	0.002214497	0.002825561
	.9	.05	0.97193670	0.04878351	(0.5667539, 1.3771195)	(0.03360175, 0.06396526)	1.034978e-04	2.959714e-08	1.438734e-03	2.432988e-05
	.5	.3	0.6023119	0.2667678	(0.4466918, 0.7579320)	(0.06487944, 0.46865612)	2.093545e-04	2.208761e-05	0.0020462381	0.0006646444

100,50	5	2	5.147613	2.034591	(3.553930, 6.741296)	(1.539917, 2.529264)	2.178961e-04	1.196507e-05	0.0014761303	0.0003459056
	2.5	7	2.566057	7.021842	(1.683882, 3.448233)	(5.494472, 8.549212)	4.363565e-05	4.770669e-06	0.0006605729	0.0002184186
	.9	.05	0.94981609	0.04820232	(0.7209524, 1.1786798)	(0.03405301, 0.06235163)	2.481643e-05	3.231666e-08	4.981609e-04	1.797683e-05
	.5	.3	0.5218649	0.2807291	(0.3609550, 0.6827748)	(0.1842645, 0.3771937)	4.780741e-06	3.713671e-06	0.0002186490	0.0001927089

Table 1 ML estimates of the parameters θ_1, θ_2 of the DG distribution, 95% Confidence interval, Mean Squared Error and Bias

In Table 1 below, n and R the sample size and number of samples, respectively. Table 1 shows that, in general, when sample size n increases, the estimates for the two parameters θ_1, θ_2 are improved. In addition, the lengths of the confidence intervals of the two parameters decrease when the sample size increases. The computed MSE and Bias for the two parameters also decrease when n increases. Therefore, one can conclude that, the results are getting better when the sample size increases. Which is true for all ML estimations; and this simulation study proves it when applying simulated data.

4. DGamma PCA of Modelling Noise

In this section, the resistance DGamma PCA for noise by demonstrates some of case study from a simulation study is performed to evaluate the DGamma PCA to the noise. Generate low rank matrices $\mathbf{B}$ 5×5 matrices from DGamma($\alpha = 9, \beta = .5$) with sample size $n=100$. Then corrupt them with noise that has rate 10% (where 10% considered the largest proportion could corrupt the data by it) and trying to recover them by DGamma PCA technique. The cases that viewed below form of 60% of cases that appeared when applying the implementation of DGamma PCA with noise.

Case 1:

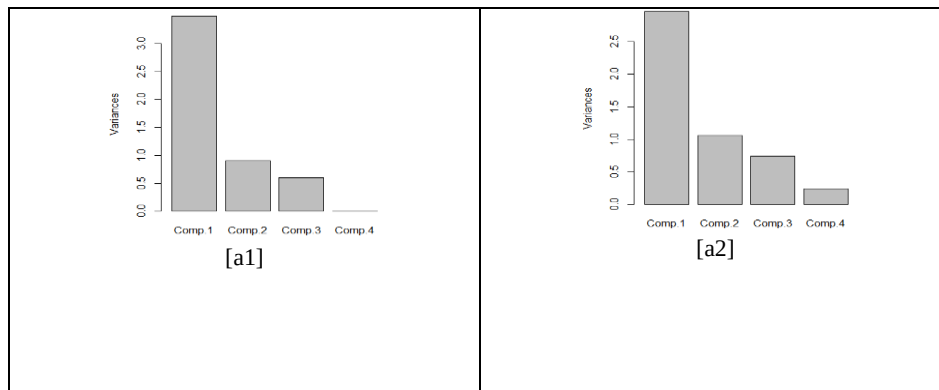


Figure 2: DGamma PCA at $n=100$; [a1] scree plot for data before adding noise [a2] scree plot when 10% noise is added.

Importance of components:				
	Comp.1	Comp.2	Comp.3	Comp.4
Standard deviation	1.86826	0.95084	0.77348	0.0848387
Proportion of Variance	0.69808	0.18082	0.11965	0.0014395
Cumulative Proportion	0.69808	0.87890	0.99856	1.0000000
[b1]				
Importance of components:				
	Comp.1	Comp.2	Comp.3	Comp.4
Standard deviation	1.72136	1.02851	0.85992	0.489464
Proportion of Variance	0.59262	0.21156	0.14789	0.047915
Cumulative Proportion	0.59262	0.80419	0.95208	1.000000
[b2]				

Table 2 Summary table of the implementation of DGamma PCA at n=100; [b1] summary of the data before nosing [b2] summary of data when noising done.

Figure 2 displays the scree plot of a simulated data comes from DGamma distribution at n=100. Firstly, from [a1] and [a2] noticing how the noise changes the variation in each component in [a2].

Secondly, from Table 2 someone can notice how the cumulative proportion was changed by comparing between [b1] and [b2]. it was in component 1 and component 2 in [b1] explain 87% of the total variation before existing any noise in data but after adding noise in data then applying DGamma PCA technique to recover data, the cumulative proportion of the component 1 and component 2 in [b2] becomes explain 80% of the total variation. Therefore, the result after applying DGamma PCA considers acceptable.

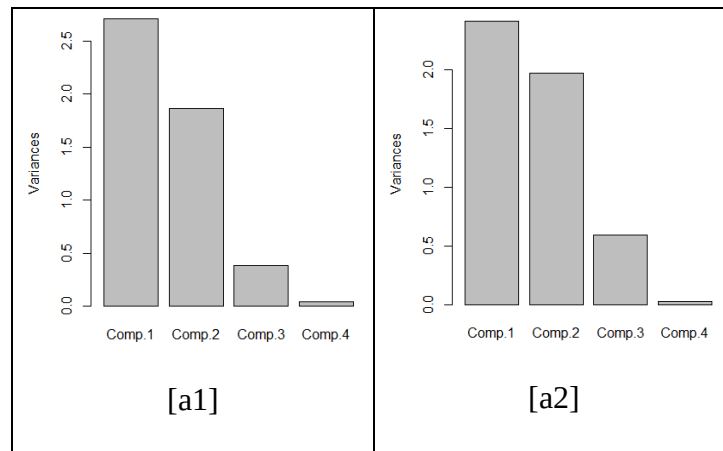
Case 2:

Figure 3 DGamma PCA at $n=100$; [a1] scree plot for data before nosing [a2] scree plot when 10% noising done.

Importance of components:

	Comp.1	Comp.2	Comp.3	Comp.4
Standard deviation	1.645290	1.364236	0.620942	0.2151970
Proportion of Variance	0.541395	0.372228	0.077113	0.0092619
Cumulative Proportion	0.541395	0.913624	0.9907380	1.000000

[b1]

Importance of components:

	Comp.1	Comp.2	Comp.3	Comp.4
Standard deviation	1.551819	1.401815	0.770416	0.1822885
Proportion of Variance	0.481628	0.393017	0.118708	0.0066458
Cumulative Proportion	0.481628	0.874645	0.993354	1.0000000

[b2]

Table 3 Summary table of the implementation of DGamma PCA at $n=100$; [b1] summary of the data before nosing [b2] summary of data when noising done.

Figure 3 displays the scree plot of a simulated data comes from DGamma distribution at $n=100$. By comparing between [a1] and [a2] will notice how the noise change the variation in each component in [a2]

In addition, from Table 3 the one can notice how the cumulative proportion was changed. The component 1 and component 2 in [b1] explain 91% of the total variation before existing any noise in data but after adding noise in data then applying DGamma PCA technique to recover data the cumulative proportion of the component 1 and component 2 in [b2] becomes explain 87% of the total variation. Therefore, DGamma PCA technique gives acceptable results.

5. Conclusions

In this paper, the DGamma distribution is viewed and some of its properties are shown. In addition, maximum likelihood estimates of the two parameters are computed and the numerical results are presented and discussed. From results, one extracts when the sample size increases the estimates results improves. Which is true for all ML estimation; and this simulation study proves it when applying simulated data. Moreover, when applying DGamma PCA technique on data with 10% noising, the results were suitable. Therefore, one can conclude that the DGamma PCA technique has acceptable behavior on data with noise.

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Received: April 19, 2018; Published: May 14, 2018